Lecture 4: Practice Problems — Answers and Hints

Math for Machine Learning

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Probability Spaces

1.1. (a) $\frac{5}{36}$. (b) $\frac{6}{36} = \frac{1}{6}$. (c) $\frac{6}{36} = \frac{1}{6}$. (d) $1 - \frac{25}{36} = \frac{11}{36}$. (e) $\frac{16-9}{36} = \frac{7}{36}$ (both dice at most 4, but not both at most 3). (f) $\frac{1}{2}$: the second die must be odd.

1.2. (a) $0.5 \cdot 0.01 + 0.3 \cdot 0.02 + 0.2 \cdot 0.03 = 0.017$. (b) $\frac{0.006}{0.017} = \frac{6}{17} \approx 0.35$ for the third machine, and also $\frac{6}{17}$ for the second; the first machine has probability $\frac{5}{17}$.

1.3. The pairs in which one number is the square of the other are $(1,1)$, $(2,4)$, $(4,2)$, $(3,9)$, $(9,3)$, so the probability is $1 - \frac{5}{100} = 0.95$.

1.4. $\frac{333+200-66}{1000} = 0.467$, and $\frac{333+200+142-66-47-28+9}{1000} = 0.543$.

1.5. $\sum_{k\geq 1} \left(\frac{5}{6}\right)^{2k-1} \frac{1}{6} = \frac{5/36}{1-25/36} = \frac{5}{11}$. Alternatively, if p is the probability, the first roll is not a six with probability $\frac{5}{6}$, and then the roles of even and odd are exchanged, so $p = \frac{5}{6}(1-p)$. The die shows a six eventually with probability 1: the event "never a six" is contained in "no six in the first n rolls" for every n, so its probability is at most $(\frac{5}{6})^n$ for every n, i.e., 0.

1.6. $P(A \cap B) = \frac{2}{6} = \frac{1}{2} \cdot \frac{4}{6}$: independent. $P(A \cap C) = \frac{1}{6} \neq \frac{1}{4}$: not independent. $P(B \cap C) = \frac{1}{2} \neq \frac{2}{3} \cdot \frac{1}{2}$: not independent.

1.7. Let C_i = "the car is behind door i" and O = "the host opens door 3". Then $P(O \mid C_1) = \frac{1}{2}$, $P(O \mid C_2) = 1$, $P(O \mid C_3) = 0$, so $P(O) = \frac{1}{2}$ and $P(C_2 \mid O) = \frac{1 \cdot 1/3}{1/2} = \frac{2}{3}$. Switching wins with probability $\frac{2}{3}$.

Random Variables and Their Distributions

2.1. (a) Binomial$(20, \frac{1}{4})$; $P(X = 5) = \binom{20}{5} \frac{3^{15}}{4^{20}} \approx 0.202$ and $P(X \geq 1) = 1 - 0.75^{20} \approx 0.9968$. (b) $\mathbb{E}[X] = 5$, $\mathrm{Var}(X) = 3.75$.

2.2. $e^{-3} \approx 0.050$; $1 - e^{-3}(1+3) \approx 0.801$; $e^{-3}\frac{3^3}{3!} \approx 0.224$.

2.3. (a) $c = \frac{3}{8}$, and $F(x) = \frac{x^3}{8}$ for $0 \leq x \leq 2$ (0 before, 1 after). (b) $P(X \leq 1) = \frac{1}{8}$, and the median is $m = 4^{1/3} \approx 1.587$.

2.4. (a) $2y$ on $[0,1]$. (b) $\frac{1}{y^2}$ for $y > 1$. (c) $\frac{1}{2}e^{-y/2}$ for $y > 0$, the Exponential$(\frac{1}{2})$ distribution.

2.5. (a) $F(x) = x^2$, so $X = \sqrt{U}$ (compare with Problem 2.4). (b) $X = (1-U)^{-1/a}$, or equivalently $U^{-1/a}$. (c) $X = \tan\left(\pi(U - \frac{1}{2})\right)$.

2.6. (a) $F = 0$ on $(-\infty, 0)$, 0.2 on $[0, 1)$, 0.7 on $[1, 3)$ and 1 on $[3, \infty)$; the jump at 1 is $0.7-0.2 = 0.5$. (b) 0, 0, 1, 1, 3 (note $F(0) = 0.2 \geq 0.2$ and $F(1) = 0.7 \geq 0.7$). (c) Return 0 if $U \leq 0.2$, 1 if $0.2 < U \leq 0.7$, and 3 if $U > 0.7$.

2.7. $\Phi(2) - \Phi(-1) = 0.9772 - 0.1587 \approx 0.819$; $1 - \Phi(1.5) \approx 0.067$; $170 \pm 1.96 \cdot 10 = [150.4, 189.6]$.

Expectation and Variance

3.1. (a) $\mathbb{E}[X] = \int_0^2 x \cdot \frac{3}{8}x^2\,dx = 1.5$, $\mathbb{E}[X^2] = 2.4$, $\mathrm{Var}(X) = 0.15$. (b) $\mathbb{E}[X] = \frac{1}{n} \cdot \frac{n(n+1)}{2}$; with $\mathbb{E}[X^2] = \frac{(n+1)(2n+1)}{6}$, $\mathrm{Var}(X) = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2-1}{12}$.

3.2. $X = \sum_i I_i$, where $I_i = 1$ if i stays at position i; $\mathbb{E}[I_i] = P(I_i = 1) = \frac{1}{n}$, so $\mathbb{E}[X] = n \cdot \frac{1}{n} = 1$. (The I_i are not independent, but linearity does not need independence.)

3.3. (a) $P(X \geq 50) \leq \frac{10}{50} = 0.2$. (b) $P(|X - 10| \geq 5) \leq \frac{4}{25} = 0.16$, and $P(X \geq 15) \leq P(|X - 10| \geq 5) \leq 0.16$.

3.4. (a) $\mathbb{E}[\sqrt{X}] = 1.5 < \sqrt{2.5} \approx 1.58$, since $\sqrt{\cdot}$ is concave. (b) Apply Jensen's inequality with the convex function $t \mapsto t^2$ to the random variable X^2.

3.5. The number of boxes is $T = \sum_{k=0}^{5} T_k$, where T_k is Geometric($\frac{6-k}{6}$) with mean $\frac{6}{6-k}$. So $\mathbb{E}[T] = \frac{6}{6} + \frac{6}{5} + \cdots + \frac{6}{1} = 14.7$.

Joint Distributions and Independence

4.1. (a) $p_X = (0.2, 0.3, 0.5)$, $p_Y = (0.4, 0.6)$; $P(Y = 1 \mid X = x) = 0.5, \frac{1}{3}, 0.8$; not independent (these differ from 0.6). (b) $\mathbb{E}[X] = 1.3$, $\mathbb{E}[Y] = 0.6$, $\mathbb{E}[XY] = 0.9$, $\mathrm{Cov}(X, Y) = 0.12$; with $\mathrm{Var}(X) = 0.61$ and $\mathrm{Var}(Y) = 0.24$, $\mathrm{Corr}(X, Y) \approx 0.31$. (c) $\mathbb{E}[X \mid Y = 0] = 1$ and $\mathbb{E}[X \mid Y = 1] = 1.5$; $0.4 \cdot 1 + 0.6 \cdot 1.5 = 1.3$.

4.2. (a) 19, $16 + 9 - 12 = 13$, and $\frac{3}{2 \cdot 3} = 0.5$. (b) $\mathrm{Var} = 4t^2 + 9(1-t)^2 + 6t(1-t)$ is minimal for $t = \frac{9-3}{4+9-6} = \frac{6}{7}$, with variance $\frac{189}{49} \approx 3.86 < 4$: combining two correlated estimates (or models) can reduce the variance below that of the better one.

4.3. (a) $\int_0^1 \int_0^1 (x + y)\,dy\,dx = 1$, and $p_X(x) = x + \frac{1}{2}$. (b) $\mathbb{E}[X] = \frac{7}{12}$; $\mathbb{E}[XY] = \frac{1}{3}$, so $\mathrm{Cov}(X, Y) = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$. (c) $p(y \mid x) = \frac{x+y}{x+1/2}$ and $\mathbb{E}[Y \mid X = x] = \frac{x/2+1/3}{x+1/2}$.

4.4. (a) The density is z on $[0, 1]$ and $2 - z$ on $[1, 2]$, so $P(X + Y \leq 1.5) = 1 - \int_{1.5}^2 (2 - z)\,dz = 1 - 0.125 = 0.875$. (b) Poisson(5), and $e^{-5} \approx 0.0067$.

4.5. $\mathbb{E}[X] = \mathbb{E}[Y] = \mathbb{E}[XY] = 0$, so $\mathrm{Cov}(X, Y) = 0$. But $P(X = 1, Y = 1) = 0 \neq \frac{1}{4} \cdot \frac{1}{4}$; in fact, $X \neq 0$ forces $Y = 0$.

4.6. $m(0) = 0.5$, $m(1) = \frac{1}{3}$, $m(2) = 0.8$. Its MSE is $\mathbb{E}[\mathrm{Var}(Y \mid X)] = 0.2 \cdot 0.25 + 0.3 \cdot \frac{2}{9} + 0.5 \cdot 0.16 \approx 0.197$, compared with $\mathrm{Var}(Y) = 0.24$ for the constant prediction $\mathbb{E}[Y] = 0.6$.

4.7. (a) $M_X(t) = \frac{1}{6}(e^t + e^{2t} + \cdots + e^{6t})$. (b) $K_X(t) = \log\lambda - \log(\lambda - t)$, $K_X'(t) = \frac{1}{\lambda - t}$, $K_X''(t) = \frac{1}{(\lambda-t)^2}$: $\mathbb{E}[X] = \frac{1}{\lambda}$, $\mathrm{Var}(X) = \frac{1}{\lambda^2}$. (c) $\left(\frac{\lambda}{\lambda - t}\right)^n$, with mean $\frac{n}{\lambda}$ and variance $\frac{n}{\lambda^2}$.

4.8. $ta - K_X(t) = ta - \lambda(e^t - 1)$ is largest at $e^t = \frac{a}{\lambda}$. For $\lambda = 10$, $a = 20$: $e^{-(20\log 2 - 10)} \approx 0.021$. Chebyshev: $P(|X - 10| \geq 10) \leq \frac{10}{100} = 0.1$.

Random Vectors and Covariance Matrices

5.1. (a) 6 and 2; the variance is largest in the direction $\frac{1}{\sqrt{2}}(1,1)$ of the eigenvector with eigenvalue 3 (the other eigenvalue is 1). (b) $\Sigma^{-1/2} = \frac{1}{2}\begin{pmatrix} 1+1/\sqrt{3} & 1/\sqrt{3}-1 \\ 1/\sqrt{3}-1 & 1+1/\sqrt{3} \end{pmatrix} \approx \begin{pmatrix} 0.789 & -0.211 \\ -0.211 & 0.789 \end{pmatrix}$, and $\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I$. (c) With $\Sigma^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$: $\sqrt{2/3} \approx 0.82$ and $\sqrt{2} \approx 1.41$. The point $(1,1)$ lies in the direction of large variance, where it is not unusual.

5.2. Mean $(1.5, 3)$; $S = \begin{pmatrix} 1.25 & 1.75 \\ 1.75 & 3.5 \end{pmatrix}$ with $\frac{1}{N}$, and $\begin{pmatrix} 5/3 & 7/3 \\ 7/3 & 14/3 \end{pmatrix}$ with $\frac{1}{N-1}$; correlation $\frac{1.75}{\sqrt{1.25 \cdot 3.5}} \approx 0.84$; the first principal component points in the direction $(0.479, 0.878)$ (eigenvalue 4.46 of S).

5.3. Mean $A(1,-1)^\top + \mathbf{b} = (0,3)^\top$; covariance $A\Sigma A^\top = \begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix}$.

5.4. The covariance matrix $(1-\rho)I + \rho\mathbf{1}\mathbf{1}^\top$ has the eigenvalues $1+2\rho$ (eigenvector $\mathbf{1}$) and $1-\rho$ (twice), so it is positive semidefinite if and only if $-\frac{1}{2} \le \rho \le 1$. For $\rho = -\frac{1}{2}$: with i.i.d. $W_1, W_2, W_3 \sim \mathcal{N}(0,1)$ and their mean $\bar{W}$, the variables $\sqrt{3/2}\,(W_i - \bar{W})$ work; they sum to 0, so $\mathrm{Var}(X_1+X_2+X_3) = 3 + 6\rho = 0$.

The Gaussian Distribution

6.1. (a) $X_1 + X_2 \sim \mathcal{N}(3, 4+3+4) = \mathcal{N}(3, 11)$, so $P(X_1+X_2 > 3) = \frac{1}{2}$. (b) $X_2 \mid X_1 = 3 \sim \mathcal{N}(2 + \frac{2}{4}(3-1),\ 3 - \frac{4}{4}) = \mathcal{N}(3, 2)$, and $X_1 \mid X_2 = 0 \sim \mathcal{N}(1 + \frac{2}{3}(0-2),\ 4 - \frac{4}{3}) = \mathcal{N}(-\frac{1}{3}, \frac{8}{3})$.

6.2. $L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}$; $\boldsymbol{\mu} + L\mathbf{z} = (3,\ 3-\sqrt{2}) \approx (3, 1.586)$.

6.3. $(X,Y) \sim \mathcal{N}\Big(\mathbf{0}, \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}\Big)$, and $X \mid Y = y \sim \mathcal{N}(\frac{2}{5}y, \frac{1}{5})$. For $y = 5$, the best estimate (in the sense of the MSE) is $\mathbb{E}[X \mid Y = 5] = 2$, not $\frac{5}{2}$: the measurement is shrunk towards the prior mean 0.

6.4. $X - Y \sim \mathcal{N}(2, 13)$, and $P(X > Y) = \Phi(\frac{2}{\sqrt{13}}) \approx \Phi(0.555) \approx 0.71$.

6.5. $\mathbb{E}[Z_i^2] = 1$ and $\mathrm{Var}(Z_i^2) = 3 - 1 = 2$; sum over i. For $d = 100$, $\|\mathbf{Z}\|^2$ is about 100 ± 14, so $\|\mathbf{Z}\|$ is typically between about 9.3 and 10.7. The density is largest at $\mathbf{0}$, but the volume near $\mathbf{0}$ is tiny; the volume of a shell of radius r grows like r^{d-1}, and most of the probability lies in a thin shell of radius about $\sqrt{d}$.

Limit Theorems and Monte Carlo Methods

7.1. (a) $\mathbb{E}[S] = 2100$, $\mathrm{Var}(S) = 600 \cdot \frac{35}{12} = 1750$, standard deviation ≈ 41.8; $P(S \ge 2200) \approx 1 - \Phi(\frac{2199.5-2100}{41.8}) = 1 - \Phi(2.38) \approx 0.009$. (b) The number of sixes has mean 100 and standard deviation $\sqrt{600 \cdot \frac{1}{6} \cdot \frac{5}{6}} \approx 9.13$, so the probability is approximately $1 - \Phi(2.14) \approx 0.016$ (the exact binomial value is 0.018).

7.2. Central limit theorem: $1.96\sqrt{0.05 \cdot 0.95/N} \le 0.005$ for $N \ge 7300$. Chebyshev: $\frac{0.0475}{N \cdot 0.005^2} \le 0.05$ for $N \ge 38\,000$. Hoeffding: $2e^{-2N \cdot 0.005^2} \le 0.05$ for $N \ge 73\,800$ (it does not use the small variance).

7.3. The integral is $\mathbb{E}[e^{-U^2}]$ for $U \sim \text{Uniform}(0,1)$, estimated by $\frac{1}{N}\sum_i e^{-u_i^2}$. $\text{Var}(e^{-U^2}) = 0.5981 - 0.7468^2 \approx 0.0404$, so the standard error is $\frac{0.201}{100} \approx 0.002$.

7.4. (a) $\bar{X}_N = X$, which is 0 or 1 with probability $\frac{1}{2}$ each, for every N: no convergence to $\frac{1}{2}$. (b) $\bar{X}_N$ has the same Cauchy distribution for every N, so it does not concentrate; averaging does not help when the mean does not exist (heavy tails).

7.5. By the central limit theorem, $\sqrt{N}(\bar{X}_N - \mu) \to \mathcal{N}(0, \sigma^2)$. With $g(x) = x^2$ and $g'(\mu) = 2\mu$, the delta method gives the variance $4\mu^2\sigma^2$. The standard error is about $\frac{2|\mu|\sigma}{\sqrt{N}} = \frac{4}{10} = 0.4$. For $\mu = 0$, $g'(0) = 0$, and the limit is the constant 0: then $\bar{X}_N^2$ fluctuates on the smaller scale $\frac{1}{N}$, and $N\bar{X}_N^2 \to \sigma^2 Z^2$ with $Z \sim \mathcal{N}(0,1)$.

7.6. (a) $M = 2$, acceptance rate $\frac{1}{2}$: draw X and U uniformly, and accept X if $U \le \frac{2X}{2} = X$. (b) $M = 1.5$ (the maximum of $6x(1-x)$, at $x = \frac{1}{2}$), acceptance rate $\frac{2}{3}$. (c) $\frac{p(\mathbf{x})}{q(\mathbf{x})} = \frac{(2\pi)^{-d/2}e^{-\|\mathbf{x}\|^2/2}}{(4\pi)^{-d/2}e^{-\|\mathbf{x}\|^2/4}} = 2^{d/2}e^{-\|\mathbf{x}\|^2/4}$, whose maximum (at $\mathbf{x} = \mathbf{0}$) is $2^{d/2}$. Acceptance rates $\frac{1}{2}$, $\frac{1}{32}$ and $\frac{1}{1024}$.

7.7. (a) $w(x) = \frac{e^{-x}}{0.2e^{-0.2x}} = 5e^{-0.8x}$, and $\hat{p} = \frac{1}{N}\sum_i \mathbf{1}_{y_i > 5}\, 5e^{-0.8y_i}$ with $y_i \sim \text{Exponential}(0.2)$. (b) $\mathbb{E}_q[\mathbf{1}_{Y>5}w(Y)^2] = \int_5^\infty \frac{e^{-2x}}{0.2e^{-0.2x}}\,dx = \frac{5}{1.8}e^{-9} \approx 3.43 \cdot 10^{-4}$, so the variance is $3.43 \cdot 10^{-4} - e^{-10} \approx 2.97 \cdot 10^{-4}$, compared with $p(1-p) \approx 6.69 \cdot 10^{-3}$ (about 22 times larger). Relative standard errors for $N = 10^4$: about 2.6% with importance sampling and 12% with plain Monte Carlo.

7.8. (a) $\mathbb{E}[\theta_k - y_{I_k} \mid \theta_k] = \theta_k - \bar{y} = f'(\theta_k)$. (b) $e_{k+1} = (1-\eta)e_k + \eta(y_{I_k} - \bar{y})$, where the second term has mean 0 and variance s^2 and is independent of e_k. The limit is $\frac{\eta^2 s^2}{1-(1-\eta)^2} = \frac{\eta s^2}{2-\eta}$. (c) $\bar{y} = 3$, $s^2 = \frac{14}{3}$, and the limit is $\frac{0.1 \cdot 14/3}{1.9} \approx 0.246$; for $\eta = 0.05$, it is about 0.120, roughly half.

Markov Chains

8.1. (a) Irreducible (all states communicate through state 2), and aperiodic, since $P_{11} > 0$. (b) $(0.375, 0.5, 0.125)$. (c) $\boldsymbol{\pi} = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$; $\pi_1 P_{12} = \frac{1}{8} = \pi_2 P_{21}$ and $\pi_2 P_{23} = \frac{1}{8} = \pi_3 P_{32}$. (d) 1, $\frac{1}{2}$, 0: the distance to $\boldsymbol{\pi}$ shrinks by the factor $\frac{1}{2}$ per step.

8.2. With $a = 0.2$ and $b = 0.3$: $\boldsymbol{\pi} = (0.6, 0.4)$, the second eigenvalue is $1 - a - b = 0.5$, and $P(X_t = 1) = 0.6 + 0.4 \cdot 0.5^t$, which is 0.65 for $t = 3$.

8.3. The degrees are $3, 2, 3, 2$, so $\boldsymbol{\pi} = (0.3, 0.2, 0.3, 0.2)$ by detailed balance. The eigenvalues of P are $1, 0, -\frac{1}{3}, -\frac{2}{3}$: the chain is aperiodic, since the triangle $1, 2, 3$ gives a cycle of odd length (the square alone is bipartite, with period 2 and the eigenvalue -1).

8.4. First-step analysis: from $i \in \{1, 2\}$, the next state is $i \pm 1$ with probability $\frac{1}{2}$ each. The equations give $h_1 = \frac{1}{3}$ and $h_2 = \frac{2}{3}$, i.e., $h_i = \frac{i}{3}$. The states 0 and 3 are absorbing ($P_{00} = P_{33} = 1$); the chain is not irreducible, and every mixture of the point masses at 0 and 3 is stationary.

8.5. $P = \begin{pmatrix} 0 & 0.5 & 0.5 \\ 1/3 & 1/6 & 0.5 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$. For example, $P_{21} = \frac{1}{2} \cdot \frac{0.2}{0.3} = \frac{1}{3}$. Then $\boldsymbol{\pi}P = (0.1 + 0.1,\ 0.1 + 0.05 + 0.15,\ 0.1 + 0.15 + 0.25) = (0.2, 0.3, 0.5)$.

8.6. The transitions are $a \to b$ (three times), $b \to a$, $a \to a$, $b \to c$, $c \to a$. So $P(b \mid a) = \frac{3}{4}$, $P(a \mid a) = \frac{1}{4}$, $P(a \mid b) = P(c \mid b) = \frac{1}{2}$, $P(a \mid c) = 1$, and all other transitions have probability 0 (for example, $c \to c$ or $a \to c$). A model that assigns probability 0 to every unseen word pair cannot handle new sentences; smoothing with pseudo-counts (as in Bayesian inference) fixes this.

8.7. (a) $P(X_1 = 1 \mid X_2 = 0) = 0.2$ and $P(X_1 = 1 \mid X_2 = 1) = 0.8$. (b) From $(0,0)$ and from $(1,0)$, the chain moves to $(0,0)$ with probability 0.8 and to $(1,0)$ with probability 0.2; from $(0,1)$ and $(1,1)$, it moves to $(0,1)$ with probability 0.2 and to $(1,1)$ with probability 0.8. For example, $\pi(0,0)P_{(0,0),(1,0)} = 0.4 \cdot 0.2 = 0.08 = 0.1 \cdot 0.8 = \pi(1,0)P_{(1,0),(0,0)}$. (c) The conditional probabilities are then 0.98 and 0.02. Each update leaves $(0,0)$ only with probability 0.02, and to reach the other likely state $(1,1)$, the chain has to pass through one of the unlikely states $(0,1)$ and $(1,0)$. So it stays near one of the two likely states for a long time, and successive samples are strongly correlated.

Parameter Estimation

9.1. (a) $\hat{\lambda} = \frac{1}{\bar{x}} = \frac{1}{3}$. (b) $\hat{\lambda} = \bar{x} = 2$. (c) $\hat{\mu} = 6$, $\hat{\sigma}^2 = 2$, $s^2 = 2.5$.

9.2. (a) The bias is $(c-1)\mu$ and the variance $c^2\frac{\sigma^2}{N}$; the MSE is minimal for $c = \frac{\mu^2}{\mu^2+\sigma^2/N}$. (b) $c = \frac{1}{2}$, with MSE $0.25 + 0.25 = 0.5$, compared with 1 for $\bar{X}_N$. The optimal c depends on the unknown μ, but the example shows that shrinking towards 0 can reduce the MSE when the noise is large compared with the signal, which is the idea behind ridge regression.

9.3. (a) $\hat{k} = \frac{4^2}{8} = 2$ and $\hat{s} = \frac{8}{4} = 2$. (b) $\hat{\theta} = 2\bar{x} = 80.5$, compared with the MLE 60 and the unbiased estimate 74. For the data $1, 2, 30$, the estimate $2\bar{x} = 22$ is smaller than the observation 30, which is impossible. (c) $\mathbb{E}[X] = \lambda$ gives $\hat{\lambda} = \bar{x}$, the MLE.

9.4. (a) $s = \frac{x}{\theta} - \frac{1-x}{1-\theta}$, so $\mathbb{E}[s] = 1 - 1 = 0$, and $\mathbb{E}[s^2] = \theta \cdot \frac{1}{\theta^2} + (1-\theta)\frac{1}{(1-\theta)^2} = \frac{1}{\theta} + \frac{1}{1-\theta}$. Also $\frac{\partial s}{\partial \theta} = -\frac{x}{\theta^2} - \frac{1-x}{(1-\theta)^2}$, with $-\mathbb{E}[\frac{\partial s}{\partial \theta}] = \frac{1}{\theta} + \frac{1}{1-\theta} = \frac{1}{\theta(1-\theta)}$. (b) $\log p = x \log \lambda - \lambda - \log x!$, $s = \frac{x}{\lambda} - 1$, and $I(\lambda) = \frac{\mathrm{Var}(X)}{\lambda^2} = \frac{1}{\lambda}$. The bound is $\frac{\lambda}{N} = 0.06$ (a standard deviation of 0.245), and the MLE $\bar{X}_N$ attains it, since $\mathrm{Var}(\bar{X}_N) = \frac{\lambda}{N}$. (c) $-\ell''(p) = \frac{k}{p^2} + \frac{N-k}{(1-p)^2}$, which is $\frac{N}{\hat{p}(1-\hat{p})}$ at $\hat{p} = \frac{k}{N}$. So $\mathrm{SE} = \sqrt{\hat{p}(1-\hat{p})/N} \approx 0.0192$, the same as before.

9.5. With $\bar{x} = 1.5$ and $\bar{y} = 4$, least squares gives the slope $\hat{w}_1 = 2.2$ and the intercept $\hat{w}_0 = 4 - 2.2 \cdot 1.5 = 0.7$; the residuals are $0.3, 0.1, -1.1, 0.7$, so $\hat{\sigma}^2 = \frac{1.8}{4} = 0.45$.

9.6. The likelihood of $\theta \geq 60$ is $1/\binom{\theta}{4}$ (and 0 for $\theta < 60$), which decreases in θ, so the MLE is 60. The unbiased estimate is $60 \cdot \frac{5}{4} - 1 = 74$. The MLE is biased since the largest observation is always at most θ, and usually smaller.

9.7. $\hat{p} = 0.18$, standard error $\sqrt{0.18 \cdot 0.82/400} \approx 0.0192$, interval $0.18 \pm 0.038 = [0.142, 0.218]$.

9.8. Maximize $\sum_k n_k \log p_k$ subject to $\sum_k p_k = 1$: $\frac{n_k}{p_k} = \lambda$ for all k, so $p_k = \frac{n_k}{\lambda}$, and the constraint gives $\lambda = N$.

9.9. (a) Unbiased, but not consistent: its distribution does not depend on N, and it is not concentrated at μ since $\sigma^2 > 0$. (b) Biased (the bias is $\frac{1}{N}$), but consistent, since the bias and the variance $\frac{\sigma^2}{N}$ tend to 0. (c) The expectation is $\frac{N\mu}{N-1}$, so it is biased (unless $\mu = 0$), but the bias $\frac{\mu}{N-1}$ and the variance $\frac{N\sigma^2}{(N-1)^2}$ tend to 0, so it is consistent. The MLE $\max_i X_i$ is consistent: its bias $-\frac{\theta}{N+1}$ and its variance, which behaves like $\frac{\theta^2}{N^2}$, tend to 0.

9.10. (a) $\log p = -\frac{1}{2}\log(2\pi v) - \frac{(x-\mu)^2}{2v}$, and $\frac{\partial^2}{\partial v^2}\log p = \frac{1}{2v^2} - \frac{(x-\mu)^2}{v^3}$, whose expectation is $\frac{1}{2v^2} - \frac{1}{v^2} = -\frac{1}{2v^2}$. (b) $\mathbb{E}[(X-\mu)^2] = v$, and $(X-\mu)^2 = vZ^2$ has the variance $v^2(\mathbb{E}[Z^4]-1) = 2v^2$, so $\mathrm{Var}(\hat{v}) = \frac{2v^2}{N} = \frac{1}{N\,I(v)}$. (c) $\frac{2v^2}{N-1} > \frac{2v^2}{N}$: not knowing μ costs about one observation (one "degree of freedom" is used to estimate μ). The difference disappears as $N \to \infty$.

9.11. (a) X_1X_2 is Bernoulli(θ^2), since $X_1X_2 = 1$ exactly if both are 1. So its mean is θ^2 and its variance $\theta^2(1-\theta^2)$. (b) Given $S = s$, all $\binom{N}{s}$ arrangements of the ones are equally likely, so $P(X_1 = X_2 = 1 \mid S = s) = \frac{s}{N} \cdot \frac{s-1}{N-1}$. By the Rao–Blackwell theorem, this estimator is unbiased (as one can also check with $\mathbb{E}[S(S-1)] = N(N-1)\theta^2$), and its variance is at most that of X_1X_2. (c) $\theta^2(1-\theta^2) = 0.09 \cdot 0.91 \approx 0.082$, more than 7 times the variance 0.011 of the Rao–Blackwellized estimator.

9.12. (a) $p(1-p)^{x-1} = \exp\left(x\log(1-p) + \log\frac{p}{1-p}\right)$: $T(x) = x$, $\eta = \log(1-p) < 0$, $h = 1$, and $A(\eta) = \log\frac{1-p}{p} = \eta - \log(1-e^{\eta})$. (b) $A'(\eta) = 1 + \frac{e^{\eta}}{1-e^{\eta}} = \frac{1}{1-e^{\eta}} = \frac{1}{p}$. (c) $\frac{1}{\hat{p}} = \bar{x}$, so $\hat{p} = \frac{1}{\bar{x}}$.

9.13. (a) $\lambda^N e^{-\lambda\sum_i x_i}$ (for $x_i \geq 0$): $\sum_i x_i$. (b) $\theta^{-N}\mathbf{1}_{\{\max_i x_i \leq \theta\}} \cdot \mathbf{1}_{\{\min_i x_i \geq 0\}}$: $\max_i x_i$. (c) $p^N(1-p)^{\sum_i x_i - N}$: $\sum_i x_i$.

Hypothesis Testing

10.1. (a) $P(S \geq 15) = \frac{1}{2^{20}}\sum_{k=15}^{20}\binom{20}{k} \approx 0.0207$, and the two-sided p-value is 0.0414. (b) Both reject at 0.05; neither rejects at 0.01. (c) 0.0207: for discrete statistics, the level of a test is usually smaller than the nominal 0.05.

10.2. Under H_0, the standard error is $\sqrt{0.1 \cdot 0.9/2000} \approx 0.0067$, so $z = \frac{0.08-0.10}{0.0067} \approx -2.98$ and the one-sided p-value is $\Phi(-2.98) \approx 0.0014$. Confidence interval: $0.08 \pm 1.96\sqrt{0.08 \cdot 0.92/2000} = [0.068, 0.092]$.

10.3. The expected counts are 300 and 100, so $\chi^2 = \frac{10^2}{300} + \frac{10^2}{100} = \frac{4}{3} \approx 1.33$. With $K - 1 = 1$ degree of freedom, χ^2_1 is the distribution of Z^2, and $P(Z^2 \geq c) = P(|Z| \geq \sqrt{c}) = 2(1 - \Phi(\sqrt{c}))$. So the p-value is $2(1 - \Phi(1.155)) \approx 0.25$, and 3 : 1 is not rejected. (The test is equivalent to the two-sided z-test for the proportion of dominant types: $z = \frac{290-300}{\sqrt{400 \cdot 0.75 \cdot 0.25}} \approx -1.155$, and $z^2 = \frac{4}{3}$.)

10.4. (a) Under H_0, $Z \sim \mathcal{N}(0,1)$, so $P(1 - \Phi(Z) \leq u) = P(Z \geq \Phi^{-1}(1-u)) = u$. Under H_1, Z is shifted to the right, and the p-values concentrate near 0. (b) $100 \cdot 0.05 = 5$.

10.5. (a) 5; $1 - 0.95^8 \approx 0.34$. (b) The threshold $0.05/8 = 0.00625$ rejects 1 hypothesis. (c) The thresholds $0.00625k$ are $0.00625, 0.0125, 0.01875, 0.025, 0.03125, \ldots$; $p_{(4)} = 0.021 \leq 0.025$, but $p_{(k)} > 0.00625k$ for $k \geq 5$, so 4 hypotheses are rejected.

10.6. The difference is $\frac{31}{3} - \frac{27}{4} = \frac{43}{12} \approx 3.58$. With the total 58, the difference for a group A with sum t is $\frac{t}{3} - \frac{58-t}{4}$, which is at least $\frac{43}{12}$ if and only if $t \geq 31$. Only $\{8, 11, 12\}$ and $\{9, 11, 12\}$ have $t \geq 31$, so the p-value is $\frac{2}{35} \approx 0.057$.

10.7. (a) The number of multisets of size 3 from 3 values, $\binom{5}{3} = 10$. (b) $\hat{\sigma}^2 = \frac{9+1+16}{3} = \frac{26}{3}$, so the bootstrap standard error is $\sqrt{26/9} \approx 1.70$, while $\frac{s}{\sqrt{N}} = \sqrt{13/3} \approx 2.08$; for small N, the bootstrap underestimates the uncertainty.

10.8. (a) Reject if $\bar{x} > \frac{1.645}{5} = 0.329$. Power: $P(\bar{X} > 0.329 \mid \mu = 0.5) = 1 - \Phi\big(5(0.329 - 0.5)\big) = \Phi(0.855) \approx 0.80$. (b) $\sqrt{N} \cdot 0.5 = 1.645 + 1.2816$ gives $N \approx 34.3$, so $N = 35$.

10.9. (a) $\frac{2e^{-2x}}{e^{-x}} = 2e^{-x}$, which decreases in x. (b) Reject if $x < c$ with $P_0(X < c) = 1 - e^{-c} = 0.05$, i.e., $c = -\log 0.95 \approx 0.0513$. The power is $1 - e^{-2c} = 1 - 0.95^2 = 0.0975$: a single observation hardly distinguishes the two hypotheses.

10.10. The likelihood ratio $\left(\frac{0.7}{0.5}\right)^S \left(\frac{0.3}{0.5}\right)^{10-S}$ increases with the number S of heads, so the tests reject for $S \geq c$. $P_0(S \geq 9) = \frac{11}{1024} \approx 0.011$ and $P_0(S \geq 8) = \frac{56}{1024} \approx 0.055 > 0.05$, so $c = 9$, with the level 0.011 and the power $0.7^{10} + 10 \cdot 0.7^9 \cdot 0.3 \approx 0.149$. The distribution of S is discrete, so not every level is attained.

Evaluating Models and Experiments

11.1. (a) 0.21, 0.026 and 0.012. The fold errors are positively correlated, since the training sets overlap heavily, so they vary less than independent estimates would, and the naive formula underestimates the uncertainty. (b) The differences are $0.02, 0.01, 0.03, 0.01, 0.02$, with mean 0.018 and standard deviation 0.0084, so the statistic is ≈ 4.8. Treating the errors as independent gives the standard error $\sqrt{0.0118^2 + 0.0097^2} \approx 0.015$ and the statistic ≈ 1.2. The paired comparison removes the variation that both models share (some folds are harder than others).

11.2. About $0.20 - 1.54 \cdot 0.02 \approx 0.169$, although all settings are equally good. Estimate the error of the selected setting on a test set that was not used for the selection, or with nested cross-validation.

11.3. (a) Pooled rate 0.05625, standard error $\sqrt{0.05625 \cdot 0.94375 \cdot \frac{2}{2400}} \approx 0.00665$, $z \approx 1.88$, two-sided p-value ≈ 0.060: not significant at 0.05. (b) $0.0125 \pm 1.96\sqrt{\frac{0.05 \cdot 0.95}{2400} + \frac{0.0625 \cdot 0.9375}{2400}} \approx [-0.0005, 0.0255]$, which contains 0. (c) $n \approx \frac{(1.96+0.84)^2 \cdot 2 \cdot 0.05625 \cdot 0.94375}{0.0125^2} \approx 5330$ per group, more than twice the users of the experiment: it was underpowered.

11.4. The event $|Z_2| > 1.96$ alone has probability 0.05, and the event $|Z_1| > 1.96$ adds the cases where only the first look is significant; since the correlation is less than 1, these cases have positive probability. Testing each look at the level 0.025 (the threshold 2.24) keeps the overall rate at most 0.05 by the Bonferroni correction; group sequential designs use better thresholds.

11.5. Same examples: $\mathrm{Var}(\bar{d}) = \frac{1+1-2 \cdot 0.8}{1000} = 0.0004$. Separate examples: $\frac{1+1}{1000} = 0.002$, five times larger. The second design would need 5000 examples per model.

11.6. (a) TP = 15, FP = 35, FN = 5, TN = 945. Accuracy 0.96, precision 0.3, recall 0.75, FPR $= \frac{35}{980} \approx 0.036$, $F_1 = \frac{30}{70} \approx 0.43$. (b) No: flagging nobody gives the accuracy 0.98, but finds no patient.

11.7. $4 + 3 + 2 = 9$ of the 12 pairs are ordered correctly: AUC = 0.75. The staircase goes up, right, up, right, up, right, right (in steps of $\frac{1}{3}$ up and $\frac{1}{4}$ right).

11.8. $2e^{-2N(0.01)^2} \leq 0.05$ gives $N \geq \frac{\log 40}{0.0002} \approx 18\,445$. For 1000 classifiers, $2000e^{-2N(0.01)^2} \leq 0.05$ gives $N \geq \frac{\log 40\,000}{0.0002} \approx 52\,984$. The reported error of the selected classifier is then optimistically biased; an independent test set is needed for it.

Bayesian Inference

12.1. Beta$(12, 4)$; mean $\frac{12}{16} = 0.75$; MAP $\frac{11}{14} \approx 0.786$; MLE 0.9; predictive probability 0.75. The prior acts like 2 extra heads and 2 extra tails (for the MAP) and pulls the estimates towards $\frac{1}{2}$.

12.2. Precision $1+\frac{4}{4}=2$, so the variance is $\frac{1}{2}$, and the mean is $\frac{0+4\cdot 3/4}{2}=1.5$. With 100 measurements: precision 26, mean $\frac{75}{26}\approx 2.88$, variance ≈ 0.038; the data dominate the prior.

12.3. (a) With $S=4$ goals in $N=4$ matches, the posterior is $\text{Gamma}(3+4, 2+4)=\text{Gamma}(7,6)$, with the mean $\frac{7}{6}\approx 1.17$; the MLE is $\hat{\lambda}=\frac{S}{N}=1$. (b) $\int_0^\infty e^{-\lambda}\frac{6^7}{6!}\lambda^6 e^{-6\lambda}\,d\lambda=\left(\frac{6}{7}\right)^7\approx 0.340$, compared with $e^{-1}\approx 0.368$.

12.4. (a) With $N=1$, $K+\sigma^2=1.25$ and $k_*=e^{-x_*^2/2}$, the formula for Gaussian process regression gives the mean $\frac{k_*}{1.25}\cdot 1=0.8\,e^{-x_*^2/2}$ and the variance $1-\frac{k_*^2}{1.25}=1-0.8\,e^{-x_*^2}$. (b) Means 0.8, 0.485, 0.108, 0.009, and variances 0.2, 0.706, 0.985, 1.000. Far away from the data, the posterior returns to the prior $\mathcal{N}(0,1)$: the observation carries no information about distant values. (c) The mean tends to 1 and the variance to 0: without noise, the posterior mean interpolates the data. With noise, the posterior mean is a compromise between the observation and the prior mean 0, as in Problem 12.2.

12.5. For $w>0$, $f'(w)=w-a+\lambda=0$ gives $w=a-\lambda$, which is valid if $a>\lambda$; similarly $w=a+\lambda$ for $a<-\lambda$. If $|a|\le\lambda$, then f is decreasing for $w<0$ and increasing for $w>0$, so $w^*=0$. Since the penalty has a kink at 0, small values of a are set exactly to 0, which makes lasso solutions sparse.

12.6. $S=(1+\frac{1+4}{0.5})^{-1}=\frac{1}{11}$ and $m=\frac{1}{0.5}\cdot\frac{1}{11}(1\cdot 1+2\cdot 3)=\frac{14}{11}\approx 1.27$. Predictive at $x=3$: $\mathcal{N}(\frac{42}{11},\frac{9}{11}+0.5)\approx\mathcal{N}(3.82,1.32)$. Ridge with $\lambda=0.5$: $\frac{7}{5+0.5}=\frac{14}{11}$.

12.7. (a) $1+xx'=\boldsymbol{\phi}(x)^\top\boldsymbol{\phi}(x')$ with $\boldsymbol{\phi}(x)=(1,x)$: the random functions are lines w_0+w_1x with $w_0,w_1\sim\mathcal{N}(0,1)$ independent. $\cos(x-x')=\boldsymbol{\phi}(x)^\top\boldsymbol{\phi}(x')$ with $\boldsymbol{\phi}(x)=(\cos x,\sin x)$: the random functions $a\cos x+b\sin x$ are sinusoids with the period 2π and a random amplitude and phase. (b) $\mathbf{u}^\top(K_1+K_2)\mathbf{u}=\mathbf{u}^\top K_1\mathbf{u}+\mathbf{u}^\top K_2\mathbf{u}\ge 0$. (It is the kernel of the sum of two independent Gaussian processes.) (c) For the points 0 and 1, $K=\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ has the eigenvalue -1; e.g., $\mathbf{u}=(1,-1)$ gives $\mathbf{u}^\top K\mathbf{u}=-2$. (d) The correlation is $e^{-x^2/(2\ell^2)}$: 0.607, 0.135 and 0.011.

12.8. After the first test, $P(D)=\frac{1}{6}$, and after the second,

$$P(D\mid{+}{+})=\frac{0.99\cdot\frac{1}{6}}{0.99\cdot\frac{1}{6}+0.05\cdot\frac{5}{6}}\approx 0.798. \qquad \text{Directly:}\quad \frac{0.01\cdot 0.99^2}{0.01\cdot 0.99^2+0.99\cdot 0.05^2}\approx 0.798.$$

12.9. y is the sum of the independent variables $f(x)\sim\mathcal{N}(0,c)$ and $\varepsilon\sim\mathcal{N}(0,\sigma^2)$. With $v=c+\sigma^2$, $\log p(y)=-\frac{y^2}{2v}-\frac{1}{2}\log v+\text{const}$, whose derivative $\frac{y^2}{2v^2}-\frac{1}{2v}$ is positive for $v<y^2$ and negative for $v>y^2$. Since $v\ge\sigma^2$, the maximum is at $v=y^2$ if $y^2>\sigma^2$, and otherwise the likelihood decreases in c. For $y=2$ and $\sigma^2=0.5$: $c=3.5$. The marginal likelihood chooses the signal variance that explains the size of the data beyond the noise; if the data are no larger than the noise, it concludes that there is no signal.

Information Theory

13.1. (a) $H(p)=\frac{1}{2}\cdot 1+\frac{1}{4}\cdot 2+2\cdot\frac{1}{8}\cdot 3=1.75$ bits, compared with 2 bits for the uniform distribution; $\text{KL}(p\,\|\,u)=2-1.75=0.25$ bits. (b) The average length is also 1.75 bits: each codeword has length $-\log_2 p_k$, so this code is optimal.

13.2. (a) $-\log 0.7\approx 0.357$. (b) $H(p,q)\approx 0.677$, $H(p)\approx 0.639$, $\text{KL}(p\,\|\,q)\approx 0.038$ (in nats).

13.3. $\log 2 + \frac{1+1}{8} - \frac{1}{2} \approx 0.443$ and $\log \frac{1}{2} + \frac{4+1}{2} - \frac{1}{2} \approx 1.307$. The second is larger because $\mathcal{N}(0,1)$ assigns very small densities to the far-out regions where $\mathcal{N}(1,4)$ has a lot of mass; the KL divergence $\mathrm{KL}(p \,\|\, q)$ heavily penalizes places where p is large but q is small.

13.4. (a) $h = -\mathbb{E}[\log(\lambda e^{-\lambda X})] = -\log\lambda + \lambda\mathbb{E}[X] = 1 - \log\lambda$. (b) The Gaussian has the entropy $\frac{1}{2}\log\frac{2\pi e}{\lambda^2} = \frac{1}{2}\log(2\pi e) - \log\lambda$, so the difference is $\frac{1}{2}\log(2\pi e) - 1 \approx 0.419$ for every λ. It is nonnegative by the maximum entropy property of the Gaussian: in fact, it is the KL divergence of the Gaussian with the same mean and variance from the exponential distribution, which is nonnegative by Gibbs' inequality.

13.5. $H(Y) = 1$ bit (since Y is again a fair bit) and $H(Y \mid X) = H(0.1, 0.9) \approx 0.469$ bits, so $I(X;Y) \approx 0.531$ bits. A perfect channel gives 1 bit, and a channel that flips with probability $\frac{1}{2}$ gives 0 bits.

13.6. With multipliers λ for $\sum_k p_k = 1$ and β for $\sum_k p_k E_k = \bar{E}$, the Lagrange condition is $-\log p_k - 1 - \lambda - \beta E_k = 0$, so $p_k \propto e^{-\beta E_k}$, and normalizing gives the softmax. The value of β is determined by the constraint on the mean energy.

Latent Variable Models and the EM Algorithm

14.1. Assign: $\{0,1\}$, $\{3,8,9,11\}$ ($J = 126$ with the old centers); update: centers 0.5 and 7.75, $J = 35.25$; assign: $\{0,1,3\}$, $\{8,9,11\}$ ($J = 18.94$); update: centers $\frac{4}{3}$ and $\frac{28}{3}$, $J = \frac{84}{9} \approx 9.33$; then nothing changes.

14.2. (a) 0.818, 0.182, 0.0006, 0.00003. (b) $\mu_1 \approx 1.18$ and $\mu_2 \approx 3.61$. k-means first assigns 2 to the second center (it is closer to 3), which gives the centers 1 and $\frac{11}{3}$; EM splits the point 2 between the two components.

14.3. The posterior probabilities of coin A are about $0.45, 0.80, 0.73, 0.35, 0.65$, and the new estimates are $\theta_A \approx 0.71$ and $\theta_B \approx 0.58$. The expected complete-data log-likelihood is $\sum_i \big(r_i(h_i \log\theta_A + (10-h_i)\log(1-\theta_A)) + (1-r_i)(h_i\log\theta_B + (10-h_i)\log(1-\theta_B))\big)$ plus a constant; setting its derivative with respect to θ_A to 0 gives the formula.

Stochastic Differential Equations

15.1. (a) $\mathcal{N}(0,3)$; $B(4) \sim \mathcal{N}(0,4)$, so $P(B(4) > 2) = P(Z > 1) \approx 0.159$. (b) $\mathrm{Cov} = \min(1,2) = 1$, so $\mathrm{Corr} = \frac{1}{\sqrt{2}} \approx 0.707$. $B(1) + B(2) = 2B(1) + (B(2) - B(1))$ is a sum of independent Gaussians, $\mathcal{N}(0, 4+1) = \mathcal{N}(0,5)$. (c) $B(2) = 1 + (B(2) - B(1)) \sim \mathcal{N}(1,1)$, since the increment is independent of $B(1)$. The vector $(B(1), B(2))$ has the covariance matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, so $B(1) \mid B(2) = 2$ is $\mathcal{N}(\frac{1}{2}\cdot 2,\ 1 - \frac{1}{2}) = \mathcal{N}(1, \frac{1}{2})$: the mean lies on the straight line from $(0,0)$ to $(2,2)$ (a "Brownian bridge").

15.2. (a) $X_1 = 1 - 0.2 + 0.5\sqrt{0.1}\cdot 0.5 \approx 0.879$ and $X_2 = 0.8 \cdot 0.879 - 0.5\sqrt{0.1} \approx 0.545$. (b) $\mathbb{E}[X_{k+1}] = 0.8\,\mathbb{E}[X_k]$, so $\mathbb{E}[X_k] = 0.8^k$, and $\mathbb{E}[X_{10}] \approx 0.107$, compared with $e^{-2} \approx 0.135$: the error of Euler's method for $x' = -2x$. (c) The SDE: $\frac{\sigma^2}{2\theta} = \frac{0.25}{4} = 0.0625$. The chain: $\frac{\sigma^2}{\theta(2-\theta h)} = \frac{0.25}{3.6} \approx 0.0694$. A stationary distribution exists if $|1 - 2h| < 1$, i.e., $0 < h < 1$.

15.3. The variance is $\sigma^2 h^{2\gamma} \cdot \frac{t}{h} = \sigma^2 t\, h^{2\gamma - 1}$. For $\gamma > \frac{1}{2}$, the noise disappears in the limit (we get an ODE); for $\gamma < \frac{1}{2}$, it explodes. Only $\gamma = \frac{1}{2}$ gives a limit with variance $\sigma^2 t$, independently of h, which is why Brownian motion has $\mathrm{Var}(B(t)) = t$ and the Euler–Maruyama noise is scaled by $\sqrt{h}$.

15.4. (a) The increments are $\sqrt{h}\,Z_k$ with i.i.d. $Z_k \sim \mathcal{N}(0,1)$, and hZ_k^2 has the mean h and the variance $h^2(\mathbb{E}[Z^4]-1) = 2h^2$. Summing n independent terms gives $\mathbb{E}[Q_n] = nh = T$ and $\mathrm{Var}(Q_n) = 2nh^2 = 2Th \to 0$, so $Q_n \to T$ by Chebyshev's inequality. (b) $|f((k+1)h) - f(kh)| = |f'(\xi_k)|h \le Ch$ with $C = \max|f'|$, so the sum is at most $nC^2h^2 = C^2Th \to 0$. Since $Q_n \to T > 0$, Brownian paths cannot be continuously differentiable.

15.5. (a) $x_{k+1} = x_k - \frac{h}{s^2}(x_k - \mu) + \sqrt{2h}\,z_k$, so $a = 1 - \frac{h}{s^2}$. (b) If $x_k - \mu \sim \mathcal{N}(0, v)$, then $x_{k+1} - \mu \sim \mathcal{N}(0, a^2v + 2h)$, and $v = a^2v + 2h$ gives $v = \frac{2h}{1-a^2} = \frac{2h}{2h/s^2 - h^2/s^4} = \frac{2s^2}{2 - h/s^2}$; the chain converges to it if $|a| < 1$, i.e., $0 < h < 2s^2$. The values give $\frac{0.5}{1.96} \approx 0.2551$ instead of 0.25; the bias vanishes as $h \to 0$. (c) Gradient descent on U, which converges to the minimizer μ. For a posterior, the minimizer of $U = -\log p(\theta \mid \mathcal{D})$ is the MAP estimate; the noise turns optimization into sampling.

15.6. (a) x_t is Gaussian as a sum of independent Gaussians. The mean is multiplied by $\sqrt{1-\beta}$ in each step, and the variance satisfies $v_t = (1-\beta)v_{t-1} + \beta$; if $v_{t-1} = 1 - (1-\beta)^{t-1}$, then $v_t = (1-\beta) - (1-\beta)^t + \beta = 1 - (1-\beta)^t$. So $x_t = (1-\beta)^{t/2}x_0 + \sqrt{1-(1-\beta)^t}\,z$ can be sampled in one step, without simulating the chain. (b) $t = 10$: mean 2.71, variance 0.183; $t = 100$: 1.09 and 0.867; $t = 500$: 0.019 and 1.000. For large t, x_t is approximately $\mathcal{N}(0,1)$ for every data point; generation starts from this distribution and runs the process backwards. (c) Euler–Maruyama gives $x_{k+1} = x_k - \frac{h}{2}x_k + \sqrt{h}\,z_k = (1 - \frac{\beta}{2})x_k + \sqrt{\beta}\,z_k$ for $h = \beta$. Its stationary variance $\frac{\sigma^2}{2\theta} = 1$ agrees with (b).

Measure-Theoretic Probability (Appendix)

16.1. (a) $\sigma(\{1,2\}) = \{\emptyset, \{1,2\}, \{3,4\}, \Omega\}$. The events $\{1\}$ and $\{2\}$ generate the σ-algebra of all unions of the pieces $\{1\}$, $\{2\}$, $\{3,4\}$, with $2^3 = 8$ elements: $\emptyset$, $\{1\}$, $\{2\}$, $\{3,4\}$, $\{1,2\}$, $\{1,3,4\}$, $\{2,3,4\}$, Ω. (b) X is measurable with respect to both, and $\sigma(X)$ is the first one. Y is measurable with respect to neither, since $\{Y = 3\} = \{3\}$ is in neither; $\sigma(Y)$ is the power set. (c) No: it contains $\{1\}$ and $\{3,4\}$, but not their union $\{1,3,4\}$.

16.2. (a) $\frac{n}{n+x}$ increases to 1 for every $x \ge 0$, so by monotone convergence, the limit is $\int_0^\infty e^{-x}\,dx = 1$. (b) The integrand tends to 0 for every x, and $2nx \le 1 + n^2x^2$ shows that it is at most $\frac{1}{2}$, an integrable bound on $[0,1]$; so the limit is 0 by dominated convergence. Exactly, the integral is $\frac{\log(1+n^2)}{2n} \to 0$. (c) $1 + \frac{5}{6} + \frac{4}{6} + \frac{3}{6} + \frac{2}{6} + \frac{1}{6} = \frac{21}{6} = 3.5$.

16.3. (a) $P(Y = 0) = P(Z \le 1) = \Phi(1) \approx 0.841$. Since this single point has positive probability, Y has no density with respect to the Lebesgue measure. With respect to $\delta_0 + \lambda$, it has the density $p(0) = \Phi(1)$ and $p(y) = \varphi(y+1)$ for $y > 0$ (0 for $y < 0$). (b) $\mathbb{E}[Y] = \int_1^\infty (z-1)\varphi(z)\,dz = \varphi(1) - (1 - \Phi(1)) \approx 0.0833$, using $\int_1^\infty z\varphi(z)\,dz = \varphi(1)$. By parts, $\int_1^\infty z^2\varphi(z)\,dz = \varphi(1) + 1 - \Phi(1)$, so $\mathbb{E}[Y^2] = 2(1 - \Phi(1)) - \varphi(1) \approx 0.0753$, and $\mathrm{Var}(Y) \approx 0.0753 - 0.0833^2 \approx 0.0684$.

16.4. (a) $\mathbb{E}[X \mid \mathcal{G}] = 3$ on $\{1,3,5\}$ and 4 on $\{2,4,6\}$; $\mathbb{E}[X^2 \mid \mathcal{G}] = \frac{35}{3}$ on $\{1,3,5\}$ and $\frac{56}{3}$ on $\{2,4,6\}$. Tower property: $\frac{1}{2}\cdot 3 + \frac{1}{2}\cdot 4 = 3.5 = \mathbb{E}[X]$. (b) $\mathrm{Var}(X \mid \mathcal{G}) = \frac{35}{3} - 9 = \frac{56}{3} - 16 = \frac{8}{3}$ on both events, and $\mathrm{Var}(\mathbb{E}[X \mid \mathcal{G}]) = \frac{1}{4}$ (the values 3 and 4 with probability $\frac{1}{2}$ each). Indeed $\frac{8}{3} + \frac{1}{4} = \frac{35}{12} = \mathrm{Var}(X)$. (c) $\mathbb{E}[(X-Z)^2] = \frac{1}{2}\mathbb{E}[(X-a)^2 \mid X \text{ even}] + \frac{1}{2}\mathbb{E}[(X-b)^2 \mid X \text{ odd}]$; each term is a quadratic function of a or b, minimal at the conditional means $a = 4$ and $b = 3$. The minimal value is $\frac{8}{3}$.

16.5. (a) The first win happens at a Geometric($\frac{1}{2}$) time τ, and $P(\tau > n) = 2^{-n} \to 0$. If $\tau = k$, the gambler has lost $1 + 2 + \cdots + 2^{k-2} = 2^{k-1} - 1$ euros and then wins 2^{k-1} euros, a total gain of

1. (b) Before the gambler stops, $M_{n+1} - M_n = \pm 2^n$ with probability $\frac{1}{2}$ each, independently of the past, so $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$; afterwards, $M_{n+1} = M_n$. Optional stopping needs a bound on the stopped martingale, but here $M_{\min(n,\tau)}$ can be as small as $-(2^n - 1)$. (So $M_n \to 1$ almost surely, but $\mathbb{E}[M_n] = 0 \not\to 1$: dominated convergence does not apply either.) (c) The debt before the win is $2^{\tau-1} - 1$, and $\mathbb{E}[2^{\tau-1}] = \sum_{k\geq 1} 2^{k-1}2^{-k} = \infty$: the expected largest debt is infinite.

16.6. (a) $\sum_n n^{-1.5} < \infty$, so by the first Borel–Cantelli lemma, almost surely only finitely many A_n happen. But $\sum_{n\geq 2} \frac{1}{n \log n} = \infty$ (compare with $\int_2^x \frac{dt}{t \log t} = \log\log x - \log\log 2$), so by the second lemma, which needs the independence, infinitely many happen almost surely. (b) $\sum_n P(X_n \neq 0) = \sum_n \frac{1}{n^2} < \infty$, so almost surely $X_n = 0$ for all large n, and $X_n \to 0$ almost surely. But $\mathbb{E}[|X_n|] = n^2 \cdot \frac{1}{n^2} = 1$, so $X_n \not\to 0$ in L^1.